

# EMOTION RECOGNITION FROM SPEECH

A Deep Learning and Speech Signal Processing Approach

## PROJECT REPORT

**Objective:** Recognize human emotions such as happy, angry, sad, and neutral from speech audio.

**Core Techniques:** MFCCs, CNN, RNN, and LSTM

**Datasets:** RAVDESS, TESS, and EMO-DB

# 1. Abstract

Emotion Recognition from Speech is an important application of artificial intelligence, machine learning, and speech signal processing. The objective of this project is to automatically identify the emotional state expressed in a person's speech. Human speech contains information not only about words and language, but also about tone, pitch, energy, rhythm, and speaking style.

The proposed system uses audio preprocessing and feature extraction techniques to convert speech recordings into numerical representations. Mel-Frequency Cepstral Coefficients (MFCCs) are particularly useful because they represent the spectral characteristics of speech in a compact form suitable for machine learning. The extracted features can then be supplied to deep learning architectures such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), or Long Short-Term Memory (LSTM) networks.

Public emotional speech datasets such as RAVDESS, TESS, and EMO-DB can be used for training and testing. The final system receives a speech recording and produces an emotion label as output. Such a system can support applications including intelligent assistants, customer-service analysis, educational software, and human-computer interaction.

## Keywords

Speech Emotion Recognition, SER, MFCC, Deep Learning, CNN, RNN, LSTM, RAVDESS, TESS, EMO-DB, Audio Processing.

## 2. Introduction

Human communication is strongly influenced by emotion. Two people may say the same sentence but communicate very different meanings through their voice. Changes in pitch, loudness, speaking rate, pauses, and spectral characteristics can indicate emotional states. Speech Emotion Recognition (SER) attempts to identify these characteristics automatically from audio recordings.

Traditional speech-processing systems mainly concentrated on recognizing spoken words. Emotion recognition extends this idea by examining how something is spoken rather than only what is spoken. This makes SER a useful research area for intelligent systems that need to interact naturally with people.

Deep learning has improved speech and audio classification because neural networks can learn complex patterns from large amounts of data. CNNs are useful for learning local patterns in spectrogram-like representations, while RNNs and LSTMs are suitable for sequential information in speech.

### 2.1 Problem Statement

The main problem is to design a system that can analyze a speech signal and classify it into several predefined emotional categories. The system should handle differences in speakers, pronunciation, recording conditions, and emotional intensity while maintaining reliable classification performance.

### 2.2 Objectives

- Collect and organize emotional speech recordings.
- Preprocess audio signals to make them suitable for analysis.
- Extract informative features such as MFCCs.
- Build a deep learning model for emotion classification.
- Evaluate the model using suitable performance metrics.
- Predict emotion from a new speech recording.

## 3. Speech Signal Processing

Speech is an acoustic signal that changes continuously with time. Before applying a machine learning model, the raw waveform is commonly processed into a more useful representation. Speech signal processing provides methods for cleaning, segmenting, and representing the signal.

### 3.1 Audio Preprocessing

- 1 Load the speech recording and convert it into a consistent audio format.
- 2 Normalize the signal so amplitude differences between recordings are reduced.
- 3 Optionally remove or reduce background noise.
- 4 Trim unnecessary silence or divide long recordings into suitable segments.
- 5 Use a consistent sampling rate when preparing the dataset.

### 3.2 Important Speech Characteristics

- **Pitch:** Related to the fundamental frequency of the speaker's voice.
- **Energy:** Represents the strength or loudness of the signal.
- **Duration:** Describes how long speech sounds or segments last.
- **Speaking rate:** Represents how quickly the speaker produces speech.
- **Spectral characteristics:** Describe how energy is distributed across frequencies.

Emotions influence these characteristics in different ways. For example, an angry utterance may have strong energy and distinctive pitch patterns, while sadness may be associated with slower speech and lower energy. These are general tendencies rather than fixed rules, so a learning system is required to identify patterns from examples.

## 4. MFCC Feature Extraction

Mel-Frequency Cepstral Coefficients, commonly called MFCCs, are among the most widely used features in speech processing. MFCCs summarize the spectral envelope of speech using a representation related to human auditory perception.

### 4.1 Main Steps in MFCC Extraction

- 1 **Pre-emphasis:** The signal may be filtered to emphasize higher-frequency components.
- 2 **Framing:** Speech is divided into short overlapping frames because speech characteristics change over time.
- 3 **Windowing:** A window function is applied to reduce discontinuities at frame boundaries.
- 4 **Fast Fourier Transform:** The time-domain signal is transformed into frequency-domain information.
- 5 **Mel filter bank:** Frequency information is passed through filters distributed according to the Mel scale.
- 6 **Log operation:** Logarithmic energies are calculated.
- 7 **Discrete Cosine Transform:** Filter-bank energies are transformed to obtain MFCC coefficients.

### 4.2 Why MFCCs Are Useful

- They provide a compact representation of speech spectral information.
- They reduce the dimensionality of raw audio information.
- They are widely supported by speech-processing libraries.
- They can be used as input features for traditional and deep learning models.

For emotion recognition, MFCCs can be calculated over consecutive frames, producing a feature matrix. This matrix can be treated as an input representation for a CNN or processed as a time sequence by an RNN or LSTM.

## 5. Deep Learning Models

Deep learning models automatically learn patterns from training data. In speech emotion recognition, the model receives extracted features or learned audio representations and predicts an emotion class.

### 5.1 Convolutional Neural Network (CNN)

A CNN learns local patterns using convolutional filters. When MFCCs are represented as a two-dimensional feature map, convolutional layers can learn useful local time-frequency patterns. Pooling layers can reduce the size of intermediate representations, followed by dense layers for classification.

### 5.2 Recurrent Neural Network (RNN)

Speech contains temporal information. RNNs are designed to process sequences and maintain information from previous time steps. This makes them suitable for analyzing sequences of acoustic features.

### 5.3 Long Short-Term Memory (LSTM)

LSTM is a type of recurrent neural network designed to handle longer-term dependencies. It uses gates to control what information should be remembered, updated, or forgotten. This can help when emotional information is distributed across multiple frames of an utterance.

### 5.4 Model Comparison

| Model | Strength                              | Typical Use                          |
|-------|---------------------------------------|--------------------------------------|
| CNN   | Learns local time-frequency patterns  | MFCC/spectrogram classification      |
| RNN   | Processes sequential information      | Frame-by-frame speech features       |
| LSTM  | Captures longer temporal dependencies | Emotion patterns across an utterance |

## 6. Datasets

A supervised emotion recognition system requires speech recordings with known emotion labels. RAVDESS, TESS, and EMO-DB are commonly used datasets for speech emotion recognition experiments.

### 6.1 RAVDESS

RAVDESS stands for Ryerson Audio-Visual Database of Emotional Speech and Song. It contains acted emotional performances and is frequently used for speech and emotion research. The dataset provides labeled emotional recordings that can be divided into training, validation, and testing subsets.

### 6.2 TESS

TESS stands for Toronto Emotional Speech Set. It contains spoken words produced with different emotional expressions. Its structured recordings make it useful for classification experiments and educational projects.

### 6.3 EMO-DB

EMO-DB, also known as the Berlin Database of Emotional Speech, is a well-known acted emotional speech dataset. It contains recordings labeled with several emotion categories and is often used as a benchmark in speech emotion recognition research.

### 6.4 Dataset Preparation

- Keep audio labels correctly associated with every recording.
- Use consistent preprocessing for all samples.
- Avoid placing recordings from the same speaker in both training and test sets when evaluating speaker-independent performance.
- Check class balance and use suitable methods when classes are uneven.
- Create training, validation, and testing sets before model training.

# 7. Proposed System Architecture

The proposed system follows a sequence of processing stages. A speech recording is first collected, then preprocessed and converted into acoustic features. The feature representation is supplied to a deep learning model, which produces the predicted emotion.

## 7.1 System Flow

| Stage                  | Description                                                  |
|------------------------|--------------------------------------------------------------|
| 1. Input Audio         | Speech recording provided by the user or dataset.            |
| 2. Preprocessing       | Normalization, noise handling, trimming, and segmentation.   |
| 3. Feature Extraction  | MFCCs and optional pitch/energy features.                    |
| 4. Feature Preparation | Padding, reshaping, or sequence preparation.                 |
| 5. Deep Learning Model | CNN, RNN, or LSTM processes the features.                    |
| 6. Classification      | Final layer estimates probabilities for emotion classes.     |
| 7. Output              | Emotion with the highest predicted probability is displayed. |

## 7.2 Example Processing Pipeline

Speech Audio → Preprocessing → MFCC Extraction → Feature Matrix → CNN/RNN/LSTM → Softmax Classification → Emotion Label

## 7.3 Output Example

Input: A short speech recording. Output: Predicted emotion = **Happy**. The model may also provide probabilities for each available emotion class.

## 8. Model Training and Evaluation

After preparing the dataset and extracting features, the deep learning model is trained using labeled examples. During training, the model adjusts its parameters to reduce classification error.

### 8.1 Training Process

- 1 Load the labeled audio dataset.
- 2 Preprocess every recording consistently.
- 3 Extract MFCC or other selected features.
- 4 Split the data into training, validation, and testing sets.
- 5 Build the CNN, RNN, or LSTM architecture.
- 6 Train the model using an appropriate optimizer and loss function.
- 7 Monitor validation performance to identify overfitting.
- 8 Evaluate the final model on unseen test data.

### 8.2 Common Evaluation Metrics

- **Accuracy:** Percentage of predictions that are correct.
- **Precision:** Measures how many predictions for a class are correct.
- **Recall:** Measures how many actual samples of a class are identified correctly.
- **F1-score:** Combines precision and recall into a single measure.
- **Confusion Matrix:** Shows correct and incorrect predictions for each class.

### 8.3 Avoiding Overfitting

Overfitting occurs when a model learns the training data too closely and performs poorly on unseen recordings. Possible methods for reducing overfitting include data augmentation, dropout, regularization, early stopping, and careful dataset splitting.

## 9. Implementation Methodology

A practical implementation can be developed using Python and commonly available audio and deep learning libraries. The exact library versions and model architecture can be selected according to project requirements.

### 9.1 Suggested Software Tools

- **Python:** Main programming language.
- **Librosa:** Loading audio and extracting MFCC and other audio features.
- **NumPy:** Numerical array processing.
- **pandas:** Dataset and metadata handling.
- **scikit-learn:** Dataset splitting and evaluation metrics.
- **TensorFlow/Keras or PyTorch:** Deep learning model development.
- **Matplotlib:** Visualization of training and evaluation results.

### 9.2 Pseudocode

#### START

Load emotional speech dataset

For each audio file:

    Load audio

    Preprocess signal

    Extract MFCC features

    Store features and emotion label

End For

Split data into training, validation, and testing sets

Build CNN/RNN/LSTM model

Train model using training data

Validate and tune model

Evaluate on test data

For new audio: preprocess → extract features → predict emotion

#### END

### 9.3 Expected Result

After successful training, the system should classify an unseen speech recording into one of the emotion categories represented in the training dataset. Performance depends on the dataset, speaker variation, recording quality, selected features, and model architecture.

## **10. Applications**

Speech emotion recognition can be integrated into systems that need to understand the affective characteristics of human speech. The technology is useful as an additional signal alongside language understanding and other contextual information.

### **10.1 Human-Computer Interaction**

Emotion-aware interfaces can use vocal characteristics to adapt responses or interaction styles. For example, an application may recognize that a user sounds frustrated and present a simpler interaction flow.

### **10.2 Customer Service**

Call-center analytics can use emotion classification as one signal for reviewing conversations and identifying interaction patterns. It can support quality analysis when used responsibly and with appropriate privacy safeguards.

### **10.3 Education**

Emotion-related speech information can potentially be used in educational systems to study engagement or interaction patterns. Such systems should avoid making high-stakes judgments based only on automated emotion predictions.

### **10.4 Virtual Assistants**

Virtual assistants can combine speech recognition, natural language understanding, and emotion-related acoustic information to create more natural interactions.

### **10.5 Entertainment**

Interactive games and multimedia applications can use speech emotion recognition to adapt scenarios or responses based on the user's vocal expression.

# 11. Advantages, Limitations and Future Scope

## 11.1 Advantages

- Can automatically classify emotional speech at scale.
- MFCCs provide compact and useful acoustic representations.
- Deep learning models can learn complex patterns from training data.
- The system can be integrated with speech-based applications.
- Multiple datasets and architectures can be compared experimentally.

## 11.2 Limitations

- Emotion is complex and cannot always be identified reliably from voice alone.
- Acted datasets may not represent natural emotional speech in real environments.
- Background noise and recording quality can affect predictions.
- Different speakers may express the same emotion differently.
- Class imbalance and limited training data can reduce generalization.
- Predictions may be sensitive to language, culture, and context.

## 11.3 Future Scope

- Use larger and more diverse real-world speech datasets.
- Combine MFCCs with pitch, energy, chroma, spectral contrast, and learned representations.
- Explore transformer-based audio models and self-supervised speech representations.
- Develop multilingual emotion recognition systems.
- Combine speech, text, and visual signals when appropriate.
- Optimize models for real-time inference on mobile or edge devices.

## 11.4 Ethical Considerations

Emotion recognition should be treated as an inference rather than a perfect measurement of a person's internal emotional state. Systems should communicate uncertainty, protect audio data, obtain appropriate consent, and avoid making important decisions about people solely from automated emotion predictions.

## 12. Conclusion

Emotion Recognition from Speech is a practical application of artificial intelligence that combines speech signal processing and deep learning. The project begins with speech audio and applies preprocessing and feature extraction techniques such as MFCCs. The resulting features can be analyzed using CNN, RNN, or LSTM models to classify emotional categories.

Datasets such as RAVDESS, TESS, and EMO-DB provide labeled emotional speech for developing and evaluating the system. A complete implementation includes data preparation, feature extraction, model training, validation, testing, and prediction on new audio.

The technology has potential applications in human-computer interaction, customer-service analytics, virtual assistants, education, and entertainment. However, emotion is highly dependent on context and individual expression, so model outputs should be interpreted carefully. Future work can improve robustness through larger datasets, multilingual data, advanced neural architectures, and multimodal approaches.

### Overall Project Flow

Audio Input → Preprocessing → MFCC Feature Extraction → Deep Learning Model → Emotion Classification → Result

## 13. References and Further Reading

The following resources and well-known research works can be consulted when developing the project and writing the final implementation report.

- 1 Livingstone, S. R., & Russo, F. A. (2018). The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS). PLOS ONE.
- 2 Pichora-Fuller, M. K., et al. Resources associated with the Toronto Emotional Speech Set (TESS).
- 3 Burkhardt, F., et al. (2005). A Database of German Emotional Speech. Interspeech.
- 4 Davis, S., & Mermelstein, P. (1980). Comparison of Parametric Representations for Monosyllabic Word Recognition.
- 5 Hochreiter, S., & Schmidhuber, J. (1997). Long Short-Term Memory.
- 6 LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep Learning. Nature.
- 7 Official documentation for Python, Librosa, scikit-learn, TensorFlow/Keras, or PyTorch.

### Suggested Project Deliverables

- Dataset preparation folder and label file.
- Audio preprocessing and MFCC extraction code.
- Model training notebook or Python program.
- Accuracy and loss graphs.
- Confusion matrix and classification report.
- Sample audio prediction interface.
- Final project report and presentation.

### Final Note

The exact number of emotion classes, dataset split, model architecture, hyperparameters, and evaluation results should be reported according to the actual experiment performed. Do not claim a specific accuracy unless it has been measured on the selected test set.